

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content overlaps only narrowly with the regulatory deployment. QUAERO's EMEA subset and Mantra-GSC's EMEA subset are the only two corpora containing genuine EMA-sourced text [\[Q33, Q34, Q40\]](#), and the dataset analysis confirms these are PIL-register drug leaflets rather than SmPC text, with QUAERO EMEA holding only ~10 segmented documents across train/dev/test (QUAERO-D1, QUAERO-D9, MANTRAGSC-D11, MANTRAGSC-D12) — corroborated by [\[WEB-3, WEB-4\]](#). The remaining majority of content is clinical case reports, research abstracts, clinical trial protocols, and spoken prescriptions [\[Q19, Q24, Q29, Q38, Q43, Q44, Q45\]](#), which differ substantially from EMA QRD-templated regulatory prose in register and formulaicity [\[WEB-2, WEB-7\]](#). While pharmaceutical vocabulary (INNs, dosages, routes) does appear across corpora (DEFT2021-D1, PXCORPUS-D1, ESSAI-D1, DEFT2020-D1), the deployment's target entity nomenclature (INN/ATC/exciipient distinctions, EMA-templated posology, contraindication qualifiers) is not realized in regulatory register at scale. Off-domain content (beekeeping, sports, history in DEFT2020; veterinary and non-EU geographies in MORFITT) further dilutes the relevant signal (DEFT2020-D13, DEFT2020-D14, MORFITT-D17, MORFITT-D18). Construct-irrelevant variance is therefore substantial, though not total.

ID	Evidence
Q19	"A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2." (p.2)
Q24	"DEFT-2021 (Grouin et al., 2021) is a subset of 275 clinical cases taken from the 2019 edition of DEFT." (p.3)
Q29	"E3C (Magnini et al., 2020) is a multilingual dataset of clinical cases annotated for the NER task." (p.3)
Q33	"The QUAERO French Medical Corpus (Névéol et al., 2014), simply referred to as QUAERO in this paper, contains annotated entities and concepts for NER tasks." (p.3)
Q34	"The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE." (p.3)
Q38	"MorFITT (Labrak et al., 2023) is a multi-label dataset annotated with medical specialties. It contains 3,624 biomedical abstracts from PMC Open Access. It has been annotated across 12 medical specialties (more detail in Appendix B.5), for a total of 5,116 annotations." (p.4)
Q40	"Mantra-GSC (Kors et al., 2015) is a multilingual dataset annotated for biomedical NER. From the 5 languages covered, we included only the French subset in this benchmark. The dataset is obtained from 3 sources which have been partitioned to be evaluated separately by 2 annotation schemes (more detail in Appendix B.6): Medline (11 classes), and EMEA and Patents (10 classes). The sources cover different types of documents (biomedical abstracts/titles, drug labels and patents). To ensure evaluation consistency, we randomly split the dataset into 3 subsets: 70% for training, 10% for validation, and 20% for testing." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)
Q44	"PxCorpus (Kocabiyikoglu et al., 2022) is a spoken language understanding dataset in the domain of medical drug prescription transcripts. It includes 4 hours (1,981 recordings) of transcribed and annotated dialogues focused on drug prescriptions. The recordings were manually transcribed and semantically annotated. The first task involves classifying the textual utterances into one of the 4 intent classes (prescribe, replace, negate, none). The second task is a NER task where each word in a sequence is classified into one of 38 classes, such as drug, dose, or mode (more detail in Appendix B.9)." (p.4)

Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)
QUAERO-D1	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg/ml); phosphate de sodium, monobasique, monohydraté; chlorure de sodium; polysorbate 80 (E433) et eau pour préparation injectable. — Pharmaceutical composition list from SmPC-like regulatory text, annotating active substance and excipients
DEFT2021-D1	clarithromycine 500 mg deux fois par jour pendant 10 jours... sulfate de quinidine 200 mg trois fois par jour — Drug names with explicit dosage and frequency — relevant to deployment NER targets
PXCORPUS-D1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Full prescription structure with drug name, dose, form, frequency, indication, duration
ESSAI-D1	avec la combinaison gemcitabine + abraxane, chez des patients avec un cancer du pancréas — Drug INN + brand name co-occurrence in trial text
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
QUAERO-D9	La Commission européenne a délivré une autorisation de mise sur le marché valide dans toute l'Union européenne pour Tysabri à Elan Pharma International Ltd, le 27 juin 2006. — EMA marketing authorization reference in regulatory-genre French
MANTRAGSC-D11	train=70, validation=10, test=20 — Test set of 20 examples is statistically insufficient for robust NER evaluation
MANTRAGSC-D12	train=70, validation=10, test=20 — Same size constraint for patent genre evaluation
DEFT2020-D13	Certains apiculteurs sélectionnent leurs reines afin de favoriser au mieux la production. — Beekeeping encyclopedic content — irrelevant to regulatory deployment
DEFT2020-D14	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Sports content — zero regulatory relevance
MORFITT-D17	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe — Canine veterinary content irrelevant to human drug labeling
MORFITT-D18	Prévalence et caractéristiques morphologiques et moléculaires de Sarcocystis bertrami chez les chevaux en Chine — Equine parasitology in China; fully outside deployment scope
WEB-2	https://learning.eupati.eu/mod/book/view.php?id=858&chapterid=770
WEB-3	QUAERO EMEA HuggingFace dataset card confirms PIL leaflet text, not SmPC text (https://huggingface.co/datasets/DrBenchmark/QUAERO)
WEB-4	QUAERO official corpus page documents EMEA/MEDLINE source genres (https://quaerofrenchmed.limsi.fr/)
WEB-7	EMA QRD-templated documents have formulaic typographic conventions (section numbering, tabular posology) that may stress sentence-splitting at boundaries (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)

Strengths

- QUAERO EMEA contains genuine EMA drug leaflet text with composition lists, posology, contraindication blocks, and EMA administrative references (QUAERO-D1, QUAERO-D6, QUAERO-D9)
- DEFT2020 Task 1 includes drug leaflet sentence pairs with contraindication phrasing, dose-time thresholds, and excipient warnings (DEFT2020-D1, DEFT2020-D2, DEFT2020-D9)
- Pharmaceutical entity vocabulary (INN, dosage, route, frequency, duration) is densely present across multiple corpora — providing concept exposure even where regulatory-register annotation is absent (DEFT2021-D1, PXCORPUS-D1, ESSAI-D1)
- Patent claim genre is represented in MANTRAGSC fr_patents [Q40] (MANTRAGSC-D7, MANTRAGSC-D8)

ID	Evidence
----	----------

Q40	“Mantra-GSC (Kors et al., 2015) is a multilingual dataset annotated for biomedical NER. From the 5 languages covered, we included only the French subset in this benchmark. The dataset is obtained from 3 sources which have been partitioned to be evaluated separately by 2 annotation schemes (more detail in Appendix B.6): Medline (11 classes), and EMEA and Patents (10 classes). The sources cover different types of documents (biomedical abstracts/titles, drug labels and patents). To ensure evaluation consistency, we randomly split the dataset into 3 subsets: 70% for training, 10% for validation, and 20% for testing.” (p.4)
QUAERO-D1	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg/ml); phosphate de sodium, monobasique, monohydraté; chlorure de sodium; polysorbate 80 (E433) et eau pour préparation injectable. — Pharmaceutical composition list from SmPC-like regulatory text, annotating active substance and excipients
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
DEFT2021-D1	clarithromycine 500 mg deux fois par jour pendant 10 jours... sulfate de quinidine 200 mg trois fois par jour — Drug names with explicit dosage and frequency — relevant to deployment NER targets
PXCORPUS-D1	primperan 10 milligrammes comprimés 1 comprimé en cas de nausée toutes les 8 heures pendant 14 jours — Full prescription structure with drug name, dose, form, frequency, indication, duration
ESSAI-D1	avec la combinaison gemcitabine + abraxane, chez des patients avec un cancer du pancréas — Drug INN + brand name co-occurrence in trial text
DEFT2020-D2	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — Drug safety warning with dose-time threshold — deployment-central entity type
QUAERO-D6	N'utilisez jamais TYSABRI [...] Si vous êtes allergique (hydpersensible) au natalizumab ou à l'un des autres composants contenus dans TYSABRI [...] Si vous avez des perturbations graves du système immunitaire (dues à une maladie, telle que leucémie ou infection à VIH...) — Full contraindication block with population qualifiers, close to EU label format
MANTRAGSC-D7	Forme posologique orale selon la revendication 1, dans laquelle ledit antagoniste opioïde libérable est la naltrexone et ledit opioïde libérable est la codéine, dans laquelle le rapport de la naltrexone sur la codéine libérable est de 0,005 : 1 à 0,044 : 1. — French patent claim with INNs and dosage ratio; patent claim register confirmed
MANTRAGSC-D8	L' utilisation selon la revendication 3, où le laxatif osmotique est du glycol de polyéthylène 3350. — Excipient/drug substance in patent claim; CHEM tag does not distinguish excipient from INN
QUAERO-D9	La Commission européenne a délivré une autorisation de mise sur le marché valide dans toute l'Union européenne pour Tysabri à Elan Pharma International Ltd, le 27 juin 2006. — EMA marketing authorization reference in regulatory-genre French
DEFT2020-D9	- Ne pas utiliser chez les personnes présentant des difficultés de déglutition en raison du risque d'inhalation bronchique et de pneumopathie lipoïde. — Contraindication safety warning in EU leaflet formulaic register

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires region-specific knowledge: EMA QRD-templated phrasing (current v10.4, Feb 2024 [WEB-7, WEB-18]), EU directive 2001/83/EC compliance language [WEB-7], INN/ATC nomenclature, EMA administrative numbering (e.g., EMEA/H/C/...). QUAERO-D9 and QUAERO-D7 show occasional appearance of these conventions but they are not systematically represented.

IC-2: The benchmark content is in French biomedical register but predominantly clinical/research, not regulatory. EMA-templated standard statements ('est contre-indiqué chez les patients...', 'la posologie recommandée est de...') [WEB-7] appear only sporadically in QUAERO EMEA and DEFT2020 drug-leaflet pairs; the dominant register is narrative clinical prose

(CAS-D8, DIAMED-D6, E3C-D4).

IC-3: Western-specific knowledge per se transfers (deployment is EU-French), but several corpora contain content from non-EU contexts: MORFITT includes Canadian (MORFITT-D13, MORFITT-D14), Japanese (MORFITT-D15), and Saudi Arabian (MORFITT-D16) studies; DiaMED is sourced from The Pan African Medical Journal, with Sub-Saharan and North African settings (DIAMED-D9, DIAMED-D11). These do not transfer cleanly to EMA/ANSM jurisdictional context.

IC-4: INSUFFICIENT DOCUMENTATION — no regional regulatory annotators were recruited; benchmark author affiliations are academic NLP and clinical medicine [Q8], not regulatory affairs. Recruiting regulatory affairs specialists for QUAERO EMEA and MANTRAGSC fr_emea re-review would be the natural mitigation.

IC-5: Documented content issues harming content validity: (a) ~10 segmented PIL documents in QUAERO EMEA insufficient for SmPC-register evaluation [WEB-3, WEB-4]; (b) heterogeneous off-domain content in DEFT2020 (DEFT2020-D13, DEFT2020-D14, DEFT2020-D15) and MORFITT (MORFITT-D17, MORFITT-D18, MORFITT-D19) introduces construct-irrelevant variance; (c) PxCorpus spoken/ASR register (PXCORPUS-D10, PXCORPUS-D12) is incompatible with formal regulatory text; (d) QUAERO sentence-splitting artifacts (QUAERO-D19, QUAERO-D20) corrupt boundary-spanning regulatory phrasing.

ID	Evidence
Q8	"LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille" (p.1)
Q34	"The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE." (p.3)
DEFT2020-D1	En raison de la présence de lactose, ce médicament est contre-indiqué en cas de galactosémie congénitale, de syndrome de malabsorption du glucose et du galactose ou de déficit en lactase. — Contraindication phrasing with excipient (lactose) — directly deployment-relevant regulatory text
E3C-D4	La patiente a déjà mené une première grossesse à terme il y a 9 ans, donnant naissance à un petit garçon en bonne santé. — Narrative patient history; typical clinical case format, not regulatory document
DIAMED-D6	Monsieur Y. O. âgé de 19 ans a été hospitalisé le 03 février 2012 pour une paraplégie évoluant depuis un mois. — Classic clinical case narrative register; entirely distinct from regulatory document genre
QUAERO-D7	EMA/H/C/122 REFLUDAN... Son principe actif est la lépirudine. — Brand name and INN both tagged CHEM, demonstrates INN extraction capability
CAS-D8	Il s'agit d'une patiente âgée de 63 ans , sans antécédents particuliers , ménopausée depuis 14 ans , célibataire , sans enfants , qui a consulté pour algies pelviennes chroniques... — Prototypical clinical case narrative register, fundamentally different from regulatory document prose
QUAERO-D9	La Commission européenne a délivré une autorisation de mise sur le marché valide dans toute l'Union européenne pour Tysabri à Elan Pharma International Ltd, le 27 juin 2006. — EMA marketing authorization reference in regulatory-genre French
DIAMED-D9	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Sub-Saharan African hospital setting; non-EU regulatory context
PXCORPUS-D10	/chet — Single-token ASR artifact with no lexical content; not regulatory text
DIAMED-D11	suture de la rupture gastrique par un surjet au polyglactin 910 (vicryl*2/0) — Trade name usage in surgical narrative; clinical protocol differs from regulatory document conventions
MANTRAGSC-D11	train=70, validation=10, test=20 — Test set of 20 examples is statistically insufficient for robust NER evaluation
PXCORPUS-D12	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — English/French code-switching and informal register unlike regulatory prose
MORFITT-D13	La région du Nord-Ouest de l'Ontario présente un taux élevé et documenté d'infections de la peau... causées par une souche de Staphylococcus aureus méthycillino-résistante d'origine communautaire — Canadian clinical context; jurisdictional mismatch with EMA/ANSM
DEFT2020-D13	Certains apiculteurs sélectionnent leurs reines afin de favoriser au mieux la production. — Beekeeping encyclopedic content — irrelevant to regulatory deployment
MORFITT-D14	En 2007 et 2008, 4 423 adultes de Calgary ont répondu à des entrevues au téléphone fixe — Canadian population study; non-EU context

DEFT2020-D14	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Sports content — zero regulatory relevance
MORFITT-D15	Évaluer la charge des maladies d'origine alimentaire au Japon... pour l'année 2011 — Japanese epidemiological study; non-EU regulatory context
DEFT2020-D15	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605), gouverne la Russie à partir de 1594... — Historical encyclopedic content — off-domain for regulatory NLP
MORFITT-D16	La présente étude transversale a étudié les attitudes de médecins internes en Arabie saoudite — Saudi Arabian medical survey; non-EU jurisdiction
MORFITT-D17	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe — Canine veterinary content irrelevant to human drug labeling
MORFITT-D18	Prévalence et caractéristiques morphologiques et moléculaires de Sarcocystis bertrami chez les chevaux en Chine — Equine parasitology in China; fully outside deployment scope
QUAERO-D19	Aucune étude clinique spécifique sur les interactions médicamenteuses n Toutefois, en raison des faibles concentrations plasmatiques du ziconotide... — Truncated mid-sentence artifact from document splitting
MORFITT-D19	Le nombre croissant de colonies d'abeilles mellifères et d'apiculteurs au Canada... lutter contre les maladies — Beekeeping disease management; unrelated to pharmaceutical regulatory deployment
QUAERO-D20	r. — Single-character fragment, sentence-splitting artifact
WEB-3	QUAERO EMEA HuggingFace dataset card confirms PIL leaflet text, not SmPC text (https://huggingface.co/datasets/DrBenchmark/QUAERO)
WEB-4	QUAERO official corpus page documents EMEA/MEDLINE source genres (https://quaerofrenchmed.limsi.fr/)
WEB-7	EMA QRD-templated documents have formulaic typographic conventions (section numbering, tabular posology) that may stress sentence-splitting at boundaries (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-18	EMA QRD v10.4 template PDF — current operational template structure (https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf)

Information Gaps

- No quantification of what proportion of QUAERO EMEA or DEFT2020 Task 1 content overlaps with current EMA QRD v10.4 template phrasing.
- No evidence on whether the benchmark's pharmaceutical vocabulary coverage transfers to current EMA SmPC vocabulary frequency distributions.

Requires Expert Verification

- Whether QUAERO EMEA's drug selection (Tysabri, Refludan, Prialt) is representative of current EMA centrally authorized product population.
- Whether MANTRAGSC fr_patents claim language matches contemporary pharmaceutical patent drafting conventions.

Remediation

Gap 1: No SmPC, CTD module, or EU-RMP text is present in any DrBenchmark dataset; QUAERO EMEA holds only ~10 segmented PIL documents [WEB-3, WEB-4].

Recommendation: Construct a supplementary corpus of French-translated EMA SmPCs (centralized procedure outputs are publicly available) covering at least the major ATC therapeutic classes, with stratified train/dev/test splits and regulatory-register-aware sentence segmentation that preserves section 4.x boundaries.

ID	Evidence
WEB-3	QUAERO EMEA HuggingFace dataset card confirms PIL leaflet text, not SmPC text (https://huggingface.co/datasets/DrBenchmark/QUAERO)
WEB-4	QUAERO official corpus page documents EMEA/MEDLINE source genres (https://quaerofrenchmed.limsi.fr/)